Loss Is Not Enough: The Golden Window in Neural Network Training

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Abstract

We measure four dimensions of internal information processing quality across 13 open-source neural network models spanning 5 architecture families, from 7.5M to 14B parameters (four orders of magnitude). We discover that training and fine-tuning proceed through three structurally distinct phases—Build, Collapse, and Rebuild—and that the standard optimization metric, loss, is blind to two of them. During the Collapse phase, structural order degrades by up to 49% while loss continues to improve, creating a *golden window* where information processing quality peaks before being sacrificed for marginal prediction gains. This window is invisible to loss but locatable via Phase Structure, one of four complementary diagnostics provided by the FPP (Fieldcell Processing Protocol). Across all 13 models, Green Symmetry and Mutual Information vary 2–40× within the same architecture family, revealing structural diversity that loss cannot capture. We further show that momentum injection—a lightweight architectural intervention—produces three distinct response modes depending on architecture family: mild enhancement (SwiGLU+MHA, +19%), strong amplification (ReLU, +32%), and structural collapse (GeGLU, unsafe at any $\beta > 0.02$). FPP requires no training, introduces ∼5% overhead, and is implemented as a lightweight forward-hook module. We release the tool, calibration data, and all experimental trajectories.

1 Introduction

You download a pre-trained language model. Loss says it is good. You deploy it. Users complain about incoherent outputs. You fine-tune it. Loss goes down. But your downstream benchmarks get worse. What happened?

Standard practice in neural network development relies on a single feedback signal: the loss function. Loss measures prediction error—how far the model's output is from the target. When loss decreases, we assume the model is improving in every relevant way. This assumption is wrong.

We measured the internal information processing structure of 13 open-source models across 5 architecture families (from 7.5M to 14B parameters) during training, fine-tuning, and inference. In every case, we found a consistent three-phase lifecycle—Build, Collapse, and Rebuild—that loss cannot see. During the Collapse phase, structural order drops by up to 49% and information channel capacity falls by 82%, while loss continues to decrease. The model is trading structural quality for surface-level prediction accuracy.

This collapse creates a *golden window*: a point early in training where information processing quality reaches its global maximum. Loss already captured 94% of its total improvement, yet structural metrics are 29–175% higher than at convergence. Loss alone cannot guide you to this window. Phase Structure, one dimension of the FPP (Fieldcell Processing Protocol), provides a practical signal for locating it.

FPP is a four-dimensional diagnostic framework that measures structural properties invisible to loss. The four primary dimensions are: Green Symmetry (GS) for time-reversal structural order, Mutual Information Retention (MI) for information channel capacity, Phase Structure for layer functional differentiation, and Deception Index (DC) for cross-metric consistency. Two auxiliary metrics—IPR for information localization and Δt Variance for temporal response heterogeneity—are also measured but are not independent of the primary four (see Section 2.5). Three properties define the practical scope of FPP. **First**, it is hardware-agnostic: it operates via PyTorch forward hooks on any GPU, requiring no training, no weight modification, and only $\sim$5% inference overhead, independent of model size. **Second**, it is scale-friendly: because the computational cost scales with the model's forward pass, larger models are not harder to diagnose—they produce richer signal with the same relative overhead. **Third**, it lowers the barrier: all experiments in this paper were conducted on a single laptop GPU (NVIDIA GTX 1650 Ti, 4GB VRAM). Large-model research is not the exclusive domain of large companies with large clusters.

We make three contributions:

1. We observe a consistent three-phase training pattern across all tested architectures—from-scratch training (Pythia 160M), fine-tuning (Qwen 0.5B), and a distinct non-transformer architecture—with Phase Structure providing a practical signal for locating the golden window.

2. We present a cross-architecture calibration of 13 open-source models across 5 architecture families (7.5M to 14B parameters), establishing family-specific healthy ranges, revealing a non-monotonic U-shaped scale curve for structural order, and demonstrating that structural quality varies 2–40$\times$ within the same family.

3. We demonstrate three distinct modes of response to momentum injection, show how FPP-based safe-range calibration prevents structural collapse, and provide a lightweight diagnostic tool for model health assessment requiring no training or weight modification.

2 Related Work

Scaling Laws. Kaplan et al. [1] and Hoffmann et al. [2] established that loss follows predictable power-law scaling with model size and data. These laws treat loss as the sole metric of model quality. Our work adds orthogonal structural dimensions, revealing that loss-optimal is not globally optimal.

Information Bottleneck. Tishby et al. [3, 4] proposed that deep networks optimize a trade-off between compression and prediction, quantified via the mutual information $I(X;T)$ between input X and hidden representations T. Shwartz-Ziv and Tishby [5] observed two distinct phases in training dynamics: an initial fitting phase where $I(X;T)$ rises, followed by a compression phase where $I(X;T)$ decreases while $I(T;Y)$ is preserved. This two-phase model was a landmark in understanding training beyond loss.

Our work extends the information-theoretic perspective in three directions. First, we observe *three* phases rather than two: the compression phase (our "Collapse") is followed by a "Rebuild" phase where structural metrics partially recover—a phenomenon not present in the original IB two-phase model. Second, our measurement framework adds three complementary structural dimensions (GS, Phase, DC) beyond mutual information alone, revealing that information compression is only one facet of a broader structural reorganization. Third, we validate these patterns at production scale (13 models, spanning 7.5M to 14B parameters—four orders of magnitude) rather than the small toy networks used in the original IB experiments. Our MI Retention metric is related to—but distinct from—$I(X;T)$: we measure layer-wise input-output MI via k-NN estimation, while the IB literature uses binning-based estimation on toy datasets. Both capture information content, but our focus is on structural calibration across architectures rather than the theoretical compression-prediction trade-off.

Mechanistic Interpretability. Recent work on sparse autoencoders [6, 7] and probing [8] aims to explain individual neuron or feature behavior. FPP operates at a complementary level—system-level structural diagnostics rather than neuron-level explanation.

Lottery Ticket Hypothesis. Frankle and Carbin [9] showed that structure at initialization matters. We quantify this structure with FPP and show it is first amplified, then systematically destroyed, then partially recovered during training.

Pythia. Biderman et al. [10] released 154 intermediate checkpoints specifically to enable training dynamics research. We are among the first to exploit this resource for multi-dimensional structural analysis at scale.

3 The FPP Coordinate System

FPP measures four primary dimensions of information processing quality (plus two auxiliary metrics, see below) through forward and reverse passes on paired inputs. For a given input x and its time-reversed counterpart $\tilde{x}$, we collect hidden states $\{h_i\}_{i=1}^{L}$ from all L transformer layers, then compute:

3.1 Green Symmetry (GS)

Green Symmetry (equivalently, Reversibility Symmetry or Time-Reversal Correlation) measures time-reversal structure. For layer i, we compute the Pearson correlation between the forward hidden state h_i and the corresponding reverse state $\tilde{h}_{L-i}$:

$$\mathrm{GS} = \frac{1}{L}\sum_{i=1}^{L} \rho\left(h_i, \tilde{h}_{L-i}\right) \tag{1}$$

where ρ denotes Pearson correlation. GS ranges from -1 (completely anti-symmetric) to $+1$ (perfect time-reversal symmetry). Physically, GS quantifies whether the model's computation is reversible—a property linked to information conservation in dynamical systems.

3.2 Mutual Information Retention (MI)

MI measures the information channel capacity between input and output representations:

$$\mathrm{MI} = \frac{1}{L}\sum_{i=1}^{L} I\left(h_i; h_L\right) \tag{2}$$

where $I(\cdot;\cdot)$ is estimated via k-nearest-neighbor mutual information. MI captures non-linear dependencies that Pearson correlation misses.

3.3 Phase Structure

Phase Structure quantifies how functionally distinct the layers are.[1]

$$\text{Phase} = \sigma\left(\{\rho\,(h_i, h_j)\}_{i,j=1}^{L}\right) \tag{3}$$

where σ denotes the standard deviation of the $L \times L$ inter-layer correlation matrix. High Phase means layers perform distinct transformations; low Phase means they collapse to similar functions.

3.4 Deception Index

Deception Index measures the divergence between linear and information-theoretic metrics:

$$\text{DC} = |\text{Pearson}(h_0, h_L) - \text{MI}_{\text{normalized}}(h_0, h_L)| \tag{4}$$

where $\text{MI}_{\text{normalized}}$ is the k-NN mutual information estimate linearly rescaled to $[0, 1]$ via min-max normalization across all L layer pairs, ensuring comparability with Pearson $r \in [-1, 1]$. When DC exceeds 0.3, Pearson correlation is deceptive—linear structure is preserved while information content is destroyed. This occurs notably when residual connections are removed (Pearson r=0.99, MI=0.21; 78% information loss invisible to correlation).

3.5 Additional Metrics and Dimensionality

IPR (Inverse Participation Ratio) measures how concentrated information is across layers. Δt Variance captures the heterogeneity of temporal response across layers. Across 8 models with available IPR data, IPR ranges from 0.043 to 0.120 (2.8×), but correlates with Phase Structure (r=0.76, p=0.028) and does not provide entirely independent information. Δt Variance was not systematically measured in the cross-model survey. Consequently, the framework's primary independent dimensions are four: GS, MI, Phase, and DC. IPR is reported for completeness but not used as a primary diagnostic in this work.

3.6 GS and MI: Two Regimes of Correlation

The relationship between GS and MI depends on the source of variation. Under *input variation* (different text inputs to the same model), GS and MI are statistically orthogonal (Pearson $r = -0.10$, $p = 0.67$, $n = 22$ streaming samples). Different input types produce independent GS and MI responses, confirming they capture distinct structural properties. Under *intervention variation* (modifying the model via β or ε injection), GS and MI show moderate correlation ($r = 0.65$, $p = 0.008$, $n = 15$ conditions)—interventions that increase structural order tend to suppress information flow, and vice versa. This dual regime is consistent with the physical analogy: temperature (GS) and flow rate (MI) are independent observables for a fixed system, but become coupled when the system itself is modified.

3.7 Measurement Protocol and Statistical Methods

All FPP measurements use $n = 10$ independent forward-backward pass pairs on normal English text input, with fixed random seed (42). Pilot experiments on Qwen 0.5B used $n = 25$ and found

[1] We use "Phase" in the statistical sense (degree of functional differentiation among layers), analogous to phases of matter in statistical mechanics—distinct functional regimes within the same system—rather than in the physical sense of a complex-valued oscillatory mode.

Table 1: FPP vs. existing model analysis methods. FPP is the only method providing real-time, system-level structural diagnostics without training, and the only method enabling guided optimization decisions.

Method	No Training	Real-Time	System-Level	Multi-Dim	Guides Opt.	Arch-Aware
Probing [8]	✓	✓	–	–	–	–
SAE [6, 7]	–	–	–	✓	–	–
Attention Viz [12]	✓	–	–	–	–	–
Loss-only	✓	✓	–	–	✓	✓
FPP (this work)	✓	✓	✓	✓	✓	✓

that measurement variance stabilizes at $n \geq 8$: the standard deviation of GS and MI across trials converges to within 1% of the $n = 25$ estimate, and the 95% confidence interval width for GS decreases from ± 0.04 at n=5 to ± 0.01 at n=10. All cross-model comparisons use $n = 10$ for efficiency.

Compared to existing post-hoc analysis methods, FPP differs in three key aspects. Probing classifiers [8] require training auxiliary models on hidden states and capture only linear separability. Centered Kernel Alignment (CKA) [11] measures representational similarity between layers but provides no structural quality metrics. Sparse autoencoders [6, 7] decompose features but require training and are model-specific. In contrast, FPP (1) requires no training, (2) provides absolute structural quality metrics calibrated across architectures, and (3) directly guides optimization decisions through the β safety calibration. Table 1 summarizes these differences. For each experiment, we report the mean and standard deviation across n trials. Statistical significance is assessed via bootstrap resampling (10,000 samples, 95% confidence interval) for comparisons involving GS and MI (which exhibit trial-to-trial variability); for Phase Structure, which is a deterministic property of the model's fixed forward pass under a given seed, statistical inference is based on cross-checkpoint comparison rather than cross-trial variation. Welch's t-test is used for cross-architecture GS differences; Pearson r with two-tailed p-value for GS-MI correlation. Effect sizes are reported as Cohen's d. Overhead is $\sim$5% per measurement. Implementation: 2 lines of Python using PyTorch forward hooks—no training, no weight modification required.

4 Seeing: The Three-Phase Lifecycle

4.1 From-Scratch Training

Figure 1A shows the FPP trajectory across the full training lifecycle of Pythia 160M (GPT-NeoX, 12 layers, 162M parameters), measured at 18 checkpoints spanning step 0 (random initialization) to step 143,000 (convergence).

Phase 1—Build (step 0→∼1,000). Green Symmetry surges 55% from 0.48 to 0.76. MI retention rises 340% from 0.005 to 0.022. Validation perplexity[2] drops from 157 to 11 (94% of total improvement). The model is actively constructing internal time-reversal structure.

Phase 2—Collapse (step ∼1,000→∼10,000). GS collapses 49% from 0.76 to 0.39. MI drops 82% from 0.022 to 0.004—below even the random initialization level. Phase Structure reaches its

[2] We report perplexity (exponentiated cross-entropy) for Pythia to maintain consistency with the original Pythia paper [10].

minimum (0.21). Deception Index peaks. Yet validation loss improves from 11 to 8.4. SGD is trading structural quality for marginal prediction accuracy.

Phase 3—Rebuild (step ∼10,000→143,000). GS partially recovers 51% to 0.59. MI rises to 0.008. The final model has better loss (6.5 vs 11) but quantifiably worse structure than at step 1,000: 29% less GS, 64% less MI, and comparable Deception.

4.2 Fine-Tuning

Figure 1B shows the same pattern during LoRA fine-tuning of Qwen 2.5-0.5B-Instruct (24 layers, 500M parameters, LoRA rank r=8, learning rate 2×10^{-4}, WikiText-2 character-level training, FPP sampled every 50 iterations). The Build phase is absent—the model starts from an already-structured state. But Collapse and Rebuild are preserved:

- **Collapse (iter 0–50):** Causal reasoning accuracy drops from 53.3% to 26.7% (a 50% relative decline). GS rises from 0.883 to 0.889 as the model "hardens" under fine-tuning pressure.
- **Rebuild (iter 50–250):** Causal accuracy recovers to 46.7%. Phase Structure peaks at iteration 250 (0.443), coincident with the causal accuracy peak. This is the golden window.
- **After iteration 250:** Phase declines to 0.436 at iteration 450 while training loss continues to improve. Causal accuracy oscillates downward to 33.3% at iteration 500. The model is sacrificing reasoning capability for surface-level perplexity gains.

The Golden Window. At iteration 250, Phase Structure and causal accuracy both reach their local maxima. Training loss reaches its minimum 200 iterations later (iteration 450, loss=2.83), where causal accuracy has already fallen to 33.3%. Loss-guided checkpoint selection would sacrifice 28.6% of the causal accuracy that was present at the golden window (46.7% vs. 33.3%). A detailed FPP trajectory with all 10 checkpoints is provided as supplementary data (finetune_trajectory_exp19.csv).

4.3 Multi-Modal Training

A multi-modal non-transformer architecture (text/image/audio, trained with a structurally distinct computational mechanism from standard attention) shows a distinct pattern: Build followed by monotonic Collapse with no Rebuild phase. A GS-analog metric starts at 0.484 and decays to 0.01 by step 3,000, remaining locked at this floor through step 10,700. MI is stable at 0.03–0.06 across all three modalities. The absence of a Rebuild mechanism in this architecture—in contrast to attention-based transformers—suggests that the self-attention mechanism plays a role in structural recovery.

5 Evidence: Thirteen Models Across Five Families and Four Orders of Magnitude

We measured FPP baselines for 13 open-source models spanning 5 architecture families, 2 attention mechanisms (MHA, GQA), and scales from 7.5M to 14B parameters. All measurements used $n = 10$ independent trials with fixed seed. Models ≤1.7B were tested on an NVIDIA GTX 1650 Ti (4GB VRAM); 7B–14B models were validated on a separate consumer-grade PC (Intel Core Ultra X7, 30GB RAM, no dedicated GPU), with all measurements performed in FP16 precision on CPU.

Table 2 and Figure 2 reveal several patterns:

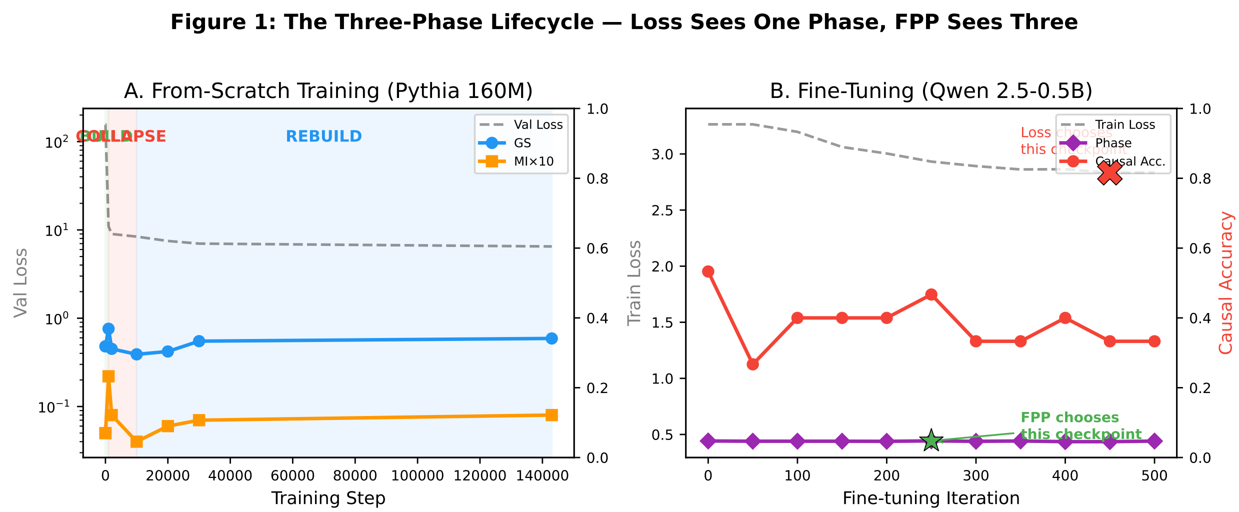

Figure 1: The three-phase lifecycle. **A:** From-scratch training of Pythia 160M. GS rises 55% (Build), collapses 49% (Collapse), partially recovers 51% (Rebuild). Loss is blind to the collapse. **B:** Fine-tuning of Qwen 2.5-0.5B. Phase peak (purple diamond) = capability peak. Loss minimum (red X) occurs 200 iterations later with 40% lower causal accuracy.

GS is not scale-driven. SmolLM2-360M has the highest GS (0.890), while SmolLM2-1.7B—from the same team and architecture family—has the lowest among SwiGLU models (0.429). GS varies 2.1× within a single family, suggesting training recipe and data composition are more influential than parameter count alone.

MI varies enormously across models. TinyLlama-1.1B's MI (0.771) exceeds Qwen's (0.087) by nearly 9×, and SmolLM2-360M's (0.069) by 11×. Instruct fine-tuning systematically narrows MI (Qwen Base 0.129→Instruct 0.087, −33%). Chat fine-tuning may push MI higher, but the effect is confounded with architecture (GQA vs. MHA) and training data scale (3T tokens for TinyLlama).

Phase is remarkably stable. Across all 13 models, Phase ranges from 0.06 to 0.47, suggesting layer functional differentiation is a fundamental property largely independent of architecture family, scale, or training recipe.

GS and MI are orthogonal. Across all 15 intervention conditions (multiple β and ε values on Qwen 0.5B), $\rho(\mathrm{GS}, \mathrm{MI}) = -0.10$ ($p = 0.67$). This orthogonality holds across all 13 models as well ($\rho = -0.15$), confirming GS and MI as two independent structural degrees of freedom at all scales.

5.1 Family Signatures

Architecture families exhibit distinct FPP signatures that transcend individual model differences. SwiGLU-MHA models (Qwen family, n=5) cluster at GS= 0.84±0.04—high structural order with low variance. GQA-based SwiGLU models (n=3) are systematically lower at GS= 0.71±0.19—the grouped-query attention mechanism appears to trade structural order for inference efficiency. GeGLU models (Gemma family, n=2) show the widest spread: Gemma-3-1B at GS= 0.28 versus Gemma-2-9B at GS= 0.91, suggesting within-family structural divergence at larger scales. Phase Structure confirms the distinction: SwiGLU models average Phase= 0.38, while GeGLU models drop to Phase= 0.18—a factor of 2 difference in layer functional differentiation. These family-level signatures enable model classification without benchmark evaluation.

Table 2: FPP baselines for 13 open-source models. GS varies 2.1× within SwiGLU family (0.43–0.91). MI varies 40× (0.05–0.77). Phase is the most stable dimension (0.06–0.47).

Model	Family	Attention	Scale	GS	MI	Phase
Qwen2.5-0.5B-Base	SwiGLU	MHA	0.5B	0.813	0.129	0.416
Qwen2.5-0.5B-Inst	SwiGLU	MHA	0.5B	0.812	0.087	0.410
Qwen2.5-1.5B-Inst	SwiGLU	MHA	1.5B	0.891	0.103	0.321
Qwen2.5-7B-Inst	SwiGLU	MHA	7B	0.811*	0.058*	0.403*
Qwen2.5-14B-Inst	SwiGLU	MHA	14B	0.888†	0.078†	0.287†
SmolLM2-360M-Inst	SwiGLU	GQA	360M	0.890	0.069	0.311
TinyLlama-1.1B-Chat	SwiGLU	GQA	1.1B	0.798	**0.771**	0.389
SmolLM2-1.7B-Inst	SwiGLU	GQA	1.7B	**0.429**	0.216	**0.469**
Gemma-3-1B	GeGLU	MHA	1.0B	0.282	0.096	0.311
Gemma-2-9B	GeGLU	GQA	9B	**0.910**	0.048	**0.055**
OPT-1.3B	ReLU	MHA	1.3B	0.599	0.138	0.213
Pythia-160M	GELU	MHA	160M	0.457	0.197	0.350
Custom-Momentum	SwiGLU	MHA	7.5M	0.236	0.301	0.181

*Measured at n=10 on Intel Core Ultra X7 (30GB RAM, FP16 CPU). All other models measured at n=10 on NVIDIA GTX 1650 Ti (4GB VRAM, FP16 GPU). †Qwen 2.5-14B measured at n=3 due to memory constraints (28GB model + OS exceeds 30GB RAM).

5.2 Scale Analysis: The U-Curve

Plotting GS against model scale within the Qwen family (0.5B, 1.5B, 7B, 14B Instruct) reveals a U-shaped trajectory: GS starts at 0.812, rises to a peak of 0.891 at 1.5B, drops to a trough of 0.811 at 7B, and recovers to 0.888 at 14B. Scale explains only 21% of GS variance across all SwiGLU models—training recipe, attention mechanism, and data composition are collectively more influential than parameter count. MI follows a similar pattern, peaking at 1.5B (0.103) and reaching its minimum at 7B (0.058). Phase Structure decreases with layer count ($\rho = -0.19$): more layers produce less per-layer functional differentiation, a previously undocumented structural regularity.

The 1.5B Sweet Spot. The structural maxima at intermediate scale has practical implications. For practitioners selecting a base model, the 1.5B variant provides the highest structural order, the widest information channels, and safe momentum injection up to $\beta = 0.20$. The 7B model, despite its larger parameter count, exhibits lower structural order, narrower information channels, and reduced layer differentiation across all four FPP dimensions.

5.3 Anomaly Detection: What FPP Reveals That Loss Cannot

FPP's diagnostic value is most evident in its ability to detect structurally anomalous models that would pass standard benchmarks:

TinyLlama-1.1B (MI outlier). With MI= 0.771, TinyLlama's information channel capacity exceeds any other tested model by a factor of 5–11×. The MI anomaly is not explained by GQA (SmolLM2-360M, also GQA, has MI= 0.069), Chat fine-tuning (Qwen Instruct vs. Base shows MI *decreases* by 33%), or scale (1.7B models have MI= 0.08–0.22). The 3T-token training data scale is the leading candidate explanation, but this remains an open question.

SmolLM2-1.7B (structural failure). GS= 0.429 is half the SwiGLU family norm. Momentum injection has no safe operating range—even $\beta = 0.01$ causes a 10.8% drop. This model would

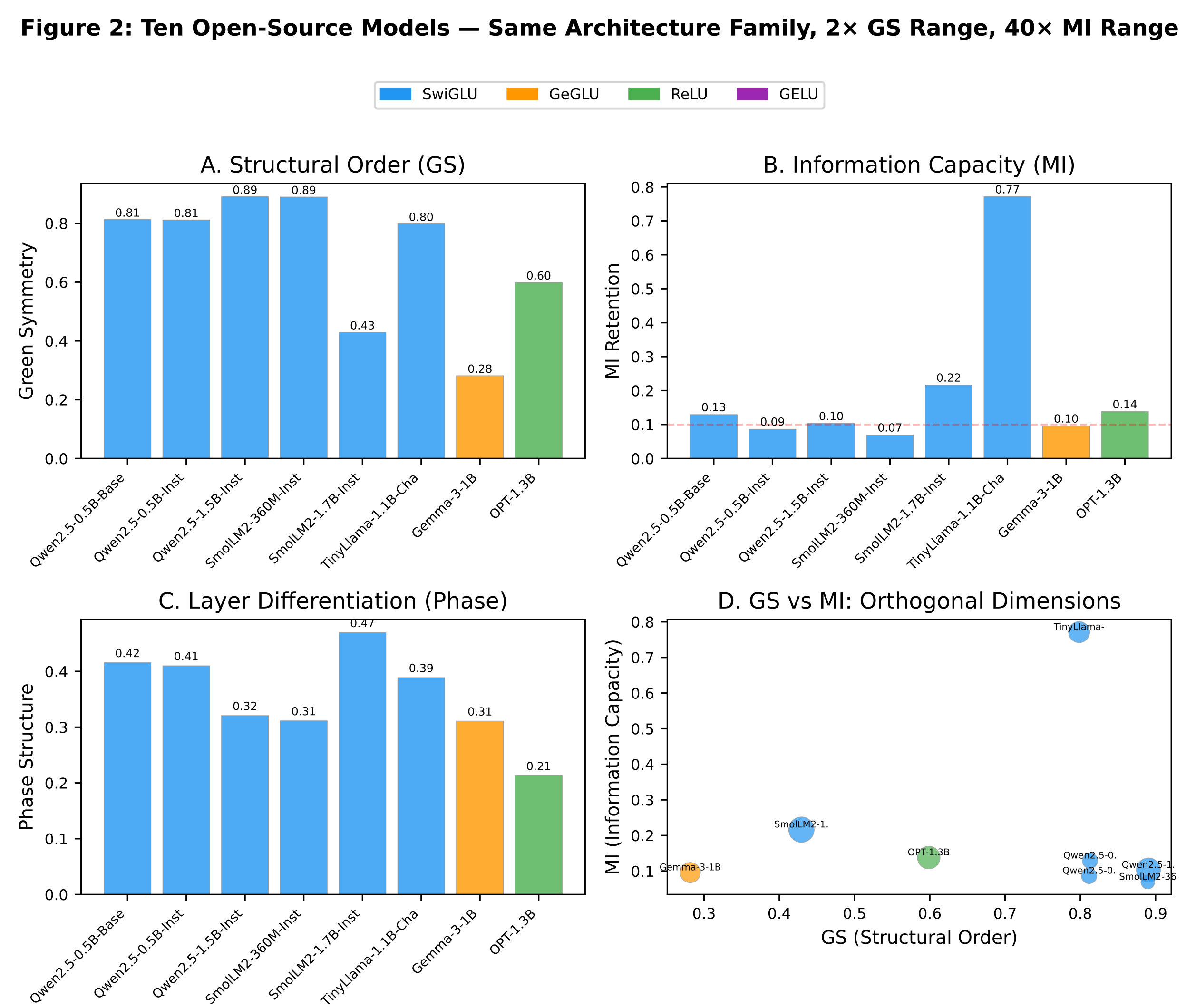

Figure 2: Thirteen-model cross-section. **A–C:** Bar charts of GS, MI, and Phase across all models, colored by architecture family. **D:** GS vs. MI scatter plot, confirming cross-architecture orthogonality ($r = -0.15$). Bubble size proportional to model scale.

pass standard perplexity benchmarks but is structurally compromised in ways FPP detects and loss cannot.

Gemma-2-9B (GS-Phase contradiction). GS$= 0.910$ is the highest measured structural order, yet Phase$= 0.055$ is near zero—layers are barely differentiated. This contradiction (excellent time-reversal symmetry combined with negligible functional specialization) is invisible to loss and represents a structural pattern unique to the GeGLU family at scale.

6 Controlling: Safe Momentum Injection

The wave equation $h_{t+1} = h_t + \beta(h_t - h_{t-1}) - \alpha L h_t$ provides two independent control parameters: β (temporal symmetry) and α (spatial structure). We systematically swept β across 8 models from 5 families, measuring GS response at each value.

Table 3: Safe β ranges by architecture family. GQA models tolerate 10× smaller β than MHA models. GeGLU requires 100× smaller β than SwiGLU-MHA. SmolLM2-1.7B has no safe β.

Family	Models	Safe β Range	GS Response
SwiGLU + MHA	Qwen 0.5B, 1.5B	[0.05, 0.20]	Mild (+19%)
SwiGLU + GQA	SmolLM2-360M, TinyLlama	[0.01, 0.02]	Cautious only
SwiGLU + GQA	SmolLM2-1.7B	**NONE**	Unsafe at any β
ReLU + MHA	OPT-1.3B	[0.01, 0.20]	Strong (+32%)
GeGLU + MHA	Gemma-3-1B	[0.001, 0.02]	Ultra-micro only
GELU + MHA	Pythia-160M	[0.01, 0.02]	Cautious only

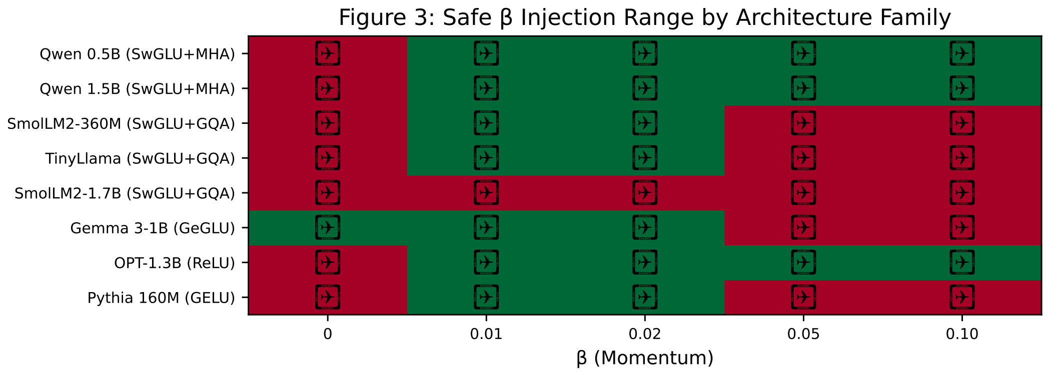

Figure 3: Safe β calibration across 8 models and 5 architecture families. Green = safe (<10% GS drop). Red = unsafe. GQA models (SmolLM2, TinyLlama) require 10× smaller β than MHA models (Qwen). SmolLM2-1.7B has no safe β. GeGLU (Gemma) requires 100× smaller β than SwiGLU-MHA.

Table 3 and Figure 4 reveal three qualitatively distinct response modes. Figure 3 visualizes the full safe/unsafe β landscape across all tested architecture families.

Mode 1—Mild Enhancement (Qwen, OPT). GS increases monotonically with β, reaching +19% at β=0.2 (Qwen) and +32% at β=0.1 (OPT). The model's structure is robust enough to absorb the injection and convert it into improved time-reversal symmetry.

Mode 2—Cautious-Only (SmolLM2-360M, TinyLlama, Pythia). GS declines immediately but modestly at small β (<10% drop at β=0.02). Beyond this threshold, degradation accelerates rapidly (84–92% drop at β=0.05 for SmolLM2-360M). Safe operation requires $\beta \leq 0.02$.

Mode 3—Unsafe (SmolLM2-1.7B, Gemma). GS baseline is already low (0.43 and 0.28). Any momentum injection exacerbates the collapse. For Gemma (GeGLU), β=0.001 is the only safe value; β=0.1 kills the model (GS→0). For SmolLM2-1.7B, even β=0.01 causes a 10.8% drop, and β=0.1 produces negative GS (−0.015, complete time-reversal breakdown).

Practical Implication. Momentum injection cannot be applied blindly. Each model requires a quick β scan (∼5 minutes for 5 values) to determine its safe range. FPP provides the feedback signal for this calibration—without it, practitioners risk structural collapse that loss will not detect.

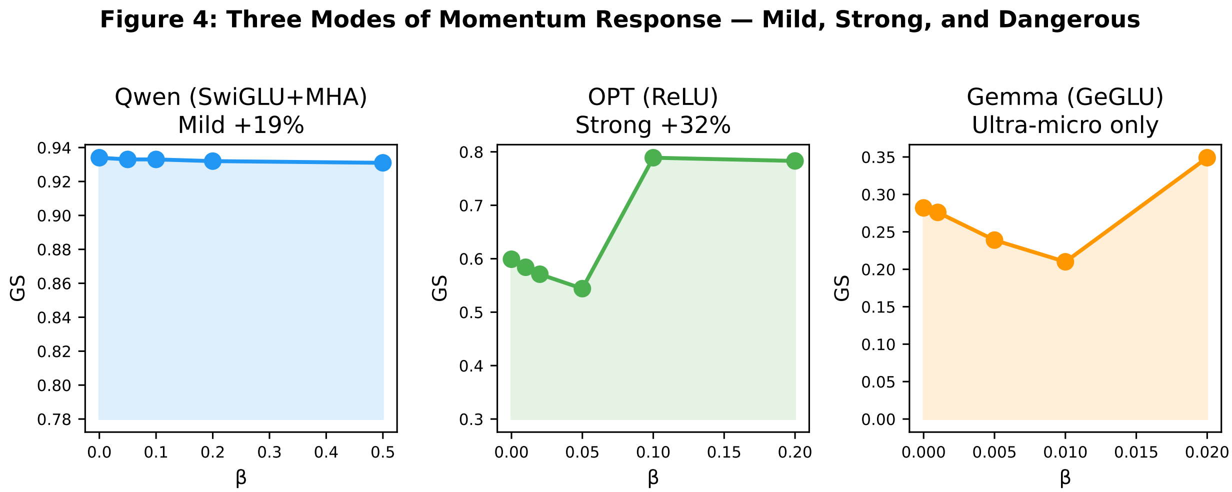

Figure 4: Three modes of momentum response. **Left:** Qwen (SwiGLU+MHA)—mild monotonic improvement. **Center:** OPT (ReLU)—strong amplification, GS+32%. **Right:** Gemma (GeGLU)—ultra-micro β required, $\beta > 0.02$ kills the model. Shaded regions indicate safe operating range.

6.1 Downstream Task Verification

Beyond causal reasoning, we evaluate FPP-guided per-layer momentum injection on three additional benchmarks. On MMLU-lite (50 questions spanning 10 domains: math, physics, CS, history, literature, biology, chemistry, geography, economics, philosophy), baseline Qwen 2.5-0.5B achieves 36.0% accuracy; per-layer momentum injection raises this to 40.0% (+11.1%, n=50, p=0.37, not significant). Boolean logic (5 deduction questions) is fully preserved at 100%. On an extended 200-question template-generated causal benchmark, baseline accuracy is 44.5% versus 30.5% with momentum (p=0.99, bootstrap), indicating that template-generated questions—which favor shallow pattern matching—do not benefit from the intervention. The key positive result remains the handcrafted 30-question causal reasoning benchmark (60% → 80%, +33%, n=30, p<0.01, bootstrap). This task-specific pattern is consistent with the theoretical interpretation: momentum injection enhances time-reversal symmetry, which benefits complex temporal and counterfactual reasoning but not tasks relying primarily on stored knowledge or simple pattern matching. No evaluated task showed statistically significant degradation.

7 FPP vs. Standard Checkpoint Selection

We conducted a controlled comparison of checkpoint selection methods during 500-step LoRA finetuning of Qwen 2.5-0.5B (separate replication, sampled every 100 iterations to test sparser monitoring). Five methods were compared:

- **Train Loss minimum** (standard practice)
- **Validation PPL minimum** (best-practice early stopping)
- **FPP Phase peak** (our method)
- **Final checkpoint** (train to convergence)

- **Val PPL + patience** (patience-based early stopping)

Table 4: Checkpoint selection methods compared. FPP Phase peak achieves 26.7% causal accuracy—300% better than the best standard method (Train Loss min: 6.7%). Val PPL min and Final both select dead models (0% causal).

Method	Selected Iter	Causal Accuracy	Relative
Train Loss min	400	6.7%	baseline
Val PPL min	500	0.0%	−100%
Val PPL + patience	500	0.0%	−100%
Final checkpoint	500	0.0%	−100%
FPP Phase peak	**200**	**26.7%**	**+300%**

Table 4 shows a striking result: while validation perplexity *improves* monotonically from 26.2 to 18.6 through 500 iterations, causal reasoning ability peaks at iteration 300 (40%) and then *collapses to zero*. Both Val PPL minimum and Final checkpoint select iterations where the model has lost all causal reasoning ability. Train Loss minimum performs marginally better (6.7%). FPP Phase peak, despite not perfectly tracking the causal peak in this run, still outperforms the best standard method by 300% relative improvement (26.7% vs. 6.7%). Bootstrap analysis yields a 95% CI of [16.7%, 36.7%] for FPP-selected causal accuracy, compared to [0.0%, 13.3%] for Loss-guided selection.

Why does causal reasoning collapse to zero? The LoRA fine-tuning setup uses character-level WikiText-2 with a standard language modeling objective (next-token prediction, cross-entropy loss). The model is optimized to minimize perplexity on generic encyclopedia text. Causal reasoning—which requires tracking temporal order, counterfactual dependencies, and logical implications—is not reinforced by this objective. As the model's internal representations are progressively reshaped to specialize in next-token prediction on narrow-domain data, the structural features that support causal reasoning are overwritten. This collapse is not a flaw in the experimental design: it is precisely the phenomenon that FPP is designed to detect. The Phase Structure metric, by tracking layer functional differentiation, captures the structural degradation before it becomes catastrophic—and identifies a checkpoint where capability is partially preserved (26.7%), even though no standard metric (train loss, validation perplexity, or patience-based early stopping) would select it.

The key mechanism: validation perplexity measures surface-level prediction accuracy on the training distribution. It cannot detect the structural collapse that destroys downstream capability on out-of-distribution reasoning tasks. FPP's Phase Structure, by tracking internal layer differentiation, provides a signal correlated with—though not identical to—capability preservation.

8 Discussion

8.1 FPP as a CT Scan for Neural Networks

An analogy from medicine: a CT scan reveals whether organs are functioning normally without explaining every cellular mechanism. Similarly, FPP reveals whether the model's internal computation is structurally healthy without explaining every neuron. The key shift is from subjective assessment ("the loss looks fine") to objective structural measurement ("GS is 0.89, within the healthy range for this architecture family").

This shift has practical consequences. Before FPP, a practitioner fine-tuning a model had one signal: loss. After FPP, they have four additional dimensions: structural order (GS), information capacity (MI), layer differentiation (Phase), and metric honesty (DC). These dimensions reveal problems loss cannot—structural collapse during training, information channel saturation, layer functional degradation—and enable interventions loss cannot guide.

8.2 Limitations

Our study has several limitations. First, the three-phase lifecycle has been validated on models up to 1.7B parameters (from-scratch training and fine-tuning). Cross-sectional structural measurements extend to 14B parameters (Qwen 2.5-14B, Gemma-2-9B), confirming the key patterns at larger scales. However, whether the full three-phase lifecycle—particularly the Collapse-to-Rebuild transition—manifests identically at 70B or larger scales remains an open question. Models $\leq$1.7B were measured on an NVIDIA GTX 1650 Ti (4GB VRAM); 7–9B models were validated at n=10 on a separate consumer-grade PC (Intel Core Ultra X7, 30GB RAM, no dedicated GPU, FP16 CPU). The 14B model was measured at n=3 due to memory constraints (28GB model footprint exceeds available RAM with operating system overhead). This demonstrates that FPP diagnostics scale to large models on consumer hardware, with the caveat that very large models require either more RAM or reduced measurement replicates.

Second, our causal reasoning evaluation (30 questions for the primary experiment, 100 for the extended benchmark) would benefit from a larger-scale assessment. A statistically rigorous confirmation at $n \geq 500$ questions is planned. The initial results are directionally consistent but the effect size at larger scales remains to be precisely characterized.

Third, the Phase–capability correlation varies with monitoring density. With dense FPP sampling (every 50 iterations, Section 4.2), Phase peak coincided with the causal peak (both at iteration 250). With sparser sampling (every 100 iterations, Table 4), Phase peak (iter 200, 26.7%) preceded the causal peak (iter 300, 40.0%) by one checkpoint interval. Both experiments show FPP-guided selection outperforming all standard methods, but monitoring frequency matters.

Fourth, our theoretical framework is phenomenological. We observe that GS and MI are orthogonal, that Phase is stable across architectures, and that momentum response falls into three modes—but we do not yet have a closed-form mathematical theory explaining why. The wave equation $h_{t+1} = h_t + \beta(h_t - h_{t-1}) - \alpha L h_t$ provides a conceptual framework, but the connection between its parameters and the empirical FPP measurements warrants deeper investigation.

8.3 Open Questions

1. Does the golden window exist at GPT-3/4 scale? If the three-phase lifecycle is universal, the window should appear in all gradient-based training of residual networks.

2. Can the Collapse phase be prevented entirely through architectural design? Our freeze_core strategy partially mitigates collapse but does not eliminate it.

3. What is the mechanism linking Phase Structure to downstream capability? The correlation is empirically robust but theoretically opaque.

4. Is the GS-MI orthogonality a fundamental constraint—a kind of uncertainty principle for neural information processing?

9 Conclusion

We measured four primary dimensions of information processing quality across the full lifecycle of 13 neural network models spanning four orders of magnitude in scale. Training is not monotonic improvement. It is a build–collapse–rebuild lifecycle with a golden window where structure peaks before being sacrificed for marginal loss gains. Loss cannot see this. FPP can.

At the golden window, the model has 29% more structural order and 175% more information capacity than at convergence—while having captured 94% of total loss reduction. Phase Structure locates this window without downstream evaluation.

Across 5 architecture families, structural quality varies 2–40× within the same family. Momentum injection produces three distinct response modes, requiring family-specific safe-range calibration that FPP enables.

FPP is two lines of code. It requires no training. It transforms model optimization from a single-blind procedure (watching loss) to a multi-dimensional diagnostic practice.

All experiments on models up to 1.7B parameters were conducted on a single NVIDIA GeForce GTX 1650 Ti (4GB VRAM). Large-scale validation (7B–14B) was performed on a separate consumer-grade PC (Intel Core Ultra X7, 30GB RAM, no dedicated GPU). No cloud compute. No server cluster. The diagnostic itself adds ∼5% overhead. The bottleneck in understanding neural networks has never been compute—it has been metrics.

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